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Applied cryptography and entropy source assessment

TECHNICAL NOTE · VERSION 5.1

NIST SP 800-90B entropy assessment as a C library

libsp80090b: measuring the min-entropy of a noise source on the target machine, deciding whether it conforms and sizing a hybrid generator

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ABSTRACT

For a physical random number generator (TRNG), assessing the entropy takes rigorous effort: modelling the noise source, capturing millions of raw samples under real operating conditions, then running long statistical tests. This effort is essential, because a deterministic random bit generator (DRBG) does not create entropy: it stretches the entropy of its seed. The security of the whole cryptographic chain therefore rests on a single quantity, the **min-entropy per sample** H of the noise source, which the **NIST SP 800-90B** recommendation [1] estimates with four tests.

libsp80090b is a C99 port of the NIST reference tool [2] as a library: its four functions assess an in-memory buffer, directly on the target machine. It gives the same values as the NIST tool, and a campaign of 52 cases on 22 datasets confirms that each test behaves as the theory predicts, with its limits: no statistical test can tell a DRBG output from real noise. The note explains the method, from the measurement to the sizing of the seed and of the health tests of a hybrid generator.

Keywords: SP 800-90B; min-entropy; IID; non-IID; restart; conditioning; health tests; hybrid generator; DRBG; SP 800-90C

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1 Introduction and context

A failing entropy source never reports its own failure. A biased, correlated or frozen source feeds the DRBG just like a good source, and the DRBG produces an output that passes every statistical test. The weakness only shows when an attacker manages to reconstruct the seed. The only protection is to **measure the min-entropy of the raw noise** before any processing, then to monitor the source continuously.

1.1 Goals of the SP 800-90B assessment

1. **Guarantee cryptographic security.** Make sure that the required entropy, for example 256 bits for the seed, is really present.
2. **Detect hardware failures.** Ageing, temperature or voltage variations, environmental disturbances.
3. **Size the generator.** Derive from H the number of samples $N = \lceil E / H \rceil$ to collect to seed the DRBG, and the cutoffs of the RCT and APT health tests.

1.2 Vocabulary

Table 1. Terms used in this document.

Term	Definition
Noise source	Non-deterministic physical or logical component: clock jitter, ring oscillator, thermal noise
Sample (symbol)	Digitised value produced by the source, 1 to 8 bits, stored one sample per byte
Raw noise	Samples before any processing or conditioning. The only legitimate subject of an assessment
Min-entropy H	$H = -\log_2(p_{max})$, where p_{max} is the probability of the most likely value: the difficulty of an adversary's best guess
IID	Independent and identically distributed samples
$H_{original}$, $H_{bitstring}$	Min-entropy of the samples as they are, and of the sequence of their bits
H_I	Initial entropy: $\min(H_{original}, n \times H_{bitstring})$ for n-bit samples
Conditioning	Deterministic function (SHA-256, HMAC, CBC-MAC) that concentrates entropy without creating any
Security strength s	Effort, in bits, to predict the DRBG output: s = 256 in our generators

2 Why a library

2.1 The need: measuring on the target machine

Entropy is not a constant of a component: it varies with the processor frequency, the load, the temperature and virtualisation. On a virtual machine, the same jitter source gave 2.67 bits per sample in one campaign, then 1.16 bits in another [10]. An assessment made once, in the laboratory, on a captured file therefore does not guarantee the value on the production machine.

2.2 Limits of the NIST reference tool

The NIST tool, SP800-90B_EntropyAssessment [2], is the reference: it produces the results of a validation submission. But it is a set of C++11 executables that work offline:

- it requires the data to be written to disk, then read back;
- it provides no API: the result has to be extracted from a text or JSON output;
- it depends on heavy libraries (libstdc++, jsoncpp, bzip2, gmp, mpfr, OpenSSL), often missing from embedded environments.

2.3 What libsp80090b provides

libsp80090b performs the four tests of the standard through **four C99 functions** that work directly on an in-memory buffer:

Non-IID assessment of a buffer of 1,000,000 samples

```
#include "sp80090b.h"

sp80090b_options_t opt = sp80090b_default_options();
sp80090b_result_t res;
opt.bits_per_symbol = 8;

/* non-IID track on 1,000,000 raw samples held in memory */
if (sp80090b_non_iid_evaluate_ex(buffer, 1000000, &opt, &res) == SP80090B_OK
    && res.warnings == 0)
    printf("Assessed min-entropy H_I = %.5f bits per sample\n", res.h_assessed);
```

Table 2. The four API functions.

Function	Test	Role
`sp80090b_iid_evaluate_ex()`	1 (§5)	Checks the IID assumption: chi-square, longest repeated substring, 10,000 permutations
`sp80090b_non_iid_evaluate_ex()`	2 (§6.3)	Computes the ten non-IID estimators
`sp80090b_restart_evaluate()`	3 (§3.1.4)	Sanity check and entropy at restart, on a 1,000 × 1,000 matrix
`sp80090b_conditioning_evaluate()`	4 (§3.1.5)	Output entropy of a conditioning function

The `_ex` suffix (*extended*) marks the version 2 functions, which take an options structure. The four labs of the validation campaign call these functions through small programs (`test_iid`, `test_non_iid`, `test_restart`, `test_conditioning`); their use is described in NT-02.

Conformity. On the eleven NIST reference datasets, the non-IID track is identical to the NIST tool to within 7.2×10^{-14}, and the IID track reproduces all of its deterministic statistics bit for bit (`make check`, NT-02).

The library does not replace three things: the **modelling** of the source, which is the designer's work; the **health tests**, which stay in the product; and the **ESV validation submission**, built with the NIST tool and assessed by NIST itself [5].

3 The SP 800-90B qualification framework

SP 800-90B splits an entropy source into three stages: the **noise source**, where the entropy comes from; the **health tests**, which check that it still works; and an optional **conditioning** stage. Figure 1 places each test in the complete chain of a generator, from the source to the SP 800-90A DRBG [12], within an SP 800-90C construction [3].

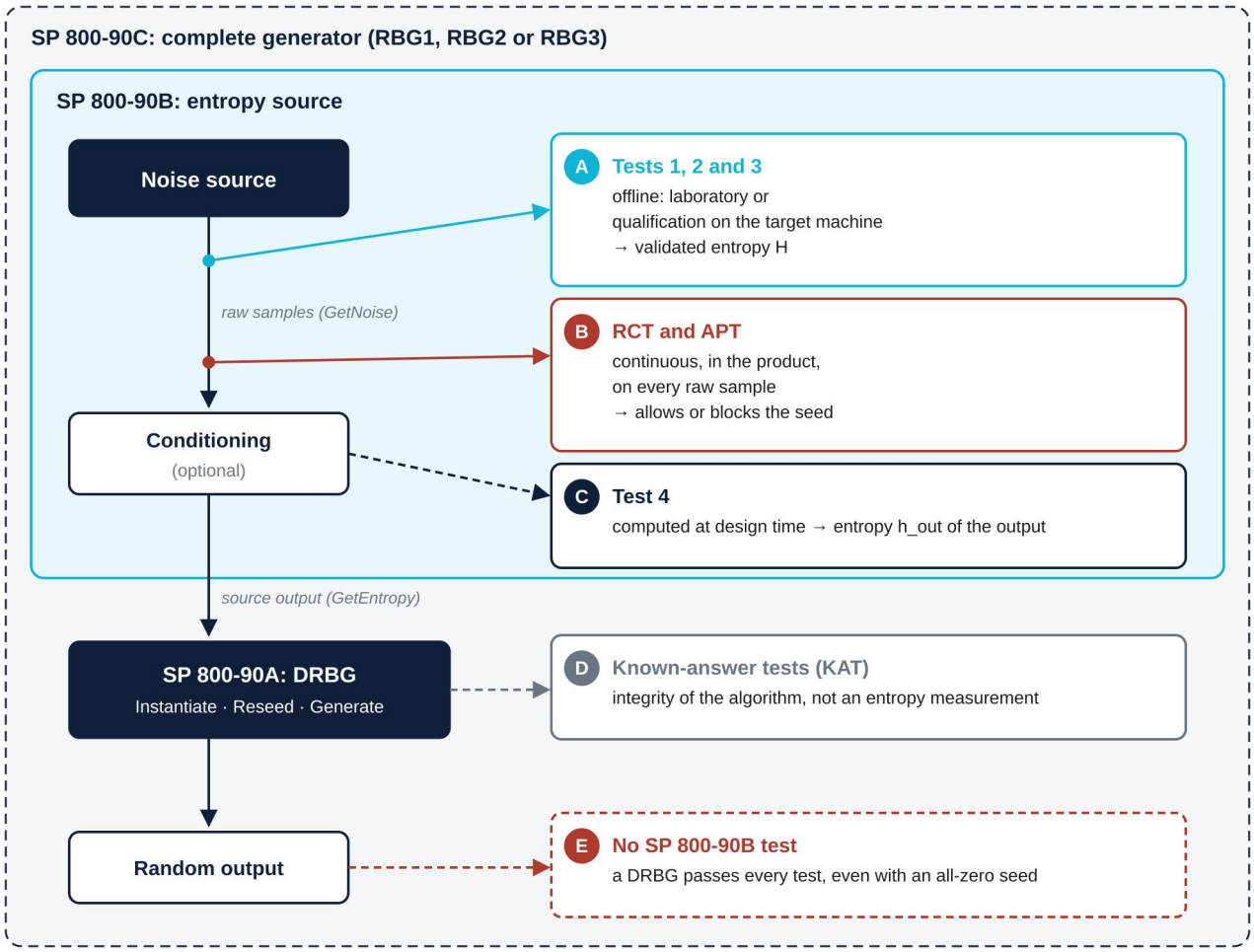


Figure 1. Place of the tests in the SP 800-90B, 90A and 90C chain: offline assessment in cyan, continuous monitoring in red.

3.1 Model first, then measure

NIST requires two complementary approaches [11]:

1. **The model.** A physical justification of the noise, which sets an entropy claim H_{claim} per sample.
2. **The statistical estimate.** A black-box check of the raw noise, performed by libsp80090b. Without knowing the inside of the source, it can be wrong: it checks the model, it does not replace it.

> **Rule.** The entropy used to size the generator is always the lower of the two: $H = \min(H_{claim}, H_{measured})$.

3.2 Offline assessment and continuous monitoring

Table 3. Offline assessment and continuous monitoring.

Criterion	Offline assessment (libsp80090b)	Continuous monitoring (RCT and APT)
Goal	Measure the min-entropy H of the source	Detect a failure of the source
Data	1,000,000 samples and 1,000 restarts	Every sample, as it is produced
When	Qualification, installation, kernel or hardware update	All the time, from start-up
Cost	Heavy: about 250 MB, from a few seconds to 2 hours of computation	Very light: a few bytes, a few operations per sample
On failure	Source refused, or claim revised downwards	Current seed discarded, output stopped, alarm

The two families complement each other: offline assessment gives the value H, continuous monitoring checks that the source does not fall below that value, since its cutoffs are computed from H.

4 The four tests of SP 800-90B

4.1 Test 1: the IID assumption (section 5)

- **Goal.** Check that the samples are independent and identically distributed. If they are, a single estimator is enough: the most common value (MCV).
- **Data.** At least 1,000,000 consecutive raw samples.
- **Method.** Two chi-square tests, independence and goodness of fit (§5.2.1 to §5.2.4); the longest repeated substring test (§5.2.5); finally 10,000 random permutations of the file (§5.1), on which 19 statistics are recomputed: excursion, increasing and decreasing runs, runs around the median, collisions, periodicity, covariance and bzip2 compression.
- **Decision.** For each statistic i, the counters $C_{i,0}$, $C_{i,1}$ and $C_{i,2}$ count the permutations whose statistic is greater than, equal to or smaller than that of the data. If $C_{i,0} + C_{i,1} \leq 5$ or $C_{i,1} + C_{i,2} \leq 5$, the IID assumption is **rejected** and the source moves to test 2.
- **Run time.** Under a minute for an IID source; 40 minutes to 2 hours for a non-IID source, which runs all 10,000 permutations.

> **Warning.** A source only follows the IID track **if its designer has justified this assumption**. Passing the IID tests does not prove independence [11]: when in doubt, apply test 2.

4.2 Test 2: the non-IID estimators (section 6.3)

- **Goal.** Estimate the min-entropy without any independence assumption: the case of almost every physical source.
- **Method.** Ten estimators, each looking for one form of regularity; four of them are predictors, which try to guess the next sample [7].

Table 4. The ten non-IID estimators.

Estimator	§	Principle	Applied to
Most common value (MCV)	6.3.1	Frequency of the most common symbol	Samples and bits
Collision	6.3.2	Mean time before a value repeats	Bits only
Markov	6.3.3	First-order Markov chain	Bits only
Compression	6.3.4	Maurer statistic on 6-bit blocks	Bits only
t-tuple	6.3.5	Frequency of the most repeated t-tuples	Samples and bits
Longest repeated substring (LRS)	6.3.6	Collisions of long substrings, through a suffix array [8]	Samples and bits
MultiMCW	6.3.7	Predictor: majority value over four sliding windows	Samples and bits
Lag	6.3.8	Predictor: value observed d positions earlier ($d \le 128$)	Samples and bits
MultiMMC	6.3.9	Predictor: Markov models of orders 1 to 16	Samples and bits
LZ78Y	6.3.10	Predictor: LZ78-type dictionary	Samples and bits

- **Result.** $H_{original}$ and $H_{bitstring}$ are the minimum of the applicable estimators; the initial entropy is $H_I = \min(H_{original}, n \times H_{bitstring})$.
- **Run time.** 5 seconds to 1 minute for 10^6 samples.

4.3 Test 3: restart (section 3.1.4)

- **Goal.** Check that the entropy measured in steady state is really present from power-up: no frozen or recurring initial state.
- **Data.** A matrix M of r = 1,000 real restarts and c = 1,000 samples per restart. Row i holds restart i; column j gathers the j-th sample of every restart.
- **Sanity check** (§3.1.4.3). The count of the most common value in each row and each column must not exceed a cutoff X_{cutoff}, obtained by simulation for an overall risk of 0.01 [6]. A source that always restarts in the same state gives a constant column and fails.
- **Validation** (§3.1.4.2). The estimates on the rows (H_r) and on the columns (H_c) must satisfy $\min(H_r, H_c) \ge H_I / 2$.
- **Result.** The validated entropy is $H = \min(H_r, H_c, H_I)$.

4.4 Test 4: conditioning (section 3.1.5)

- **Goal.** Compute the entropy h_{out} left after a conditioning function, for example SHA-256.
- **Vetted function** (HMAC, CMAC, CBC-MAC, SHA-2, SHA-3, Hash_df). h_{out} is computed with the Output_Entropy(n_{in}, n_{out}, nw, h_{in}) formula of the standard, from the input entropy h_{in}.
- **Non-vetted function.** h_{out} is further capped at $0.999 \times n_{out}$ and at $h' \times n_{out}$, where h' is the entropy per bit measured on one million outputs: $h_{out} = \min(\text{Output_Entropy}, 0.999\, n_{out}, h'\, n_{out})$.
- **Full entropy** (FIPS 140-3 IG D.K [4], SP 800-90C [3]). An output has full entropy if h_{out} equals n_{out} to within $\varepsilon \le 2^{-32}$; this requires a vetted function and a margin of at least 64 bits: $\mathbf{h_{in} \ge n_{out} + 64}$.

5 Deciding whether a source conforms

A source is declared conforming if the following five criteria are met:

1. **Data** (§3.1.1). At least 1,000,000 consecutive raw samples, with more than one symbol: a stuck source is rejected outright.

2. **Initial entropy H_I.** Through the IID track if it is justified and the assumption holds, otherwise through the ten non-IID estimators.
3. **Restart sanity.** $X_{max} \leq X_{cutoff}$ on every row and column.
4. **Restart entropy.** $\min(H_r, H_c) \geq H_I / 2$; the validated entropy is $H = \min(H_r, H_c, H_I)$.
5. **Claim.** H is at least equal to the claim H_{claim} of the model; the lower of the two values is then credited.

6 Integration into a hybrid generator

A hybrid generator combines an entropy source, assessed according to SP 800-90B, with a DRBG that stretches its seed: this is the RBG2 construction of SP 800-90C [3]. For a security strength s = 256 bits, sizing takes three steps.

6.1 Entropy to supply to the DRBG

Table 5. Entropy to supply to the DRBG for a 256-bit strength.

Operation	Required entropy E	Reason
Instantiation	384 bits (3s/2)	s bits of entropy and s/2 bits of nonce
Reseed	256 bits (s)	Restores the security strength after a compromise of the state
Full-entropy output (RBG3)	320 bits (s + 64) per 256-bit block	64-bit margin of FIPS 140-3 IG D.K

6.2 Number of raw samples

From the validated entropy H, output of test 3, the number of samples to collect is $N = \lceil E / H \rceil$.

Table 6. Number of raw samples per operation.

Source and entropy credited	H (bit/sample)	Instantiation 384	Reseed 256	Full entropy 320
jitterentropy [9], design claim (osr = 3)	0.3333	1,152	768	960
Thermal noise of Lab 3, validated H (NT-03)	4.0927	94	63	79

6.3 Health test cutoffs

The cutoffs of the health tests are computed from the credited entropy H, for a false alarm rate $\alpha = 2^{-20}$:

- **RCT** (Repetition Count Test, §4.4.1): the test fails if the same value repeats $C = 1 + \lceil 20 / H \rceil$ times in a row;
- **APT** (Adaptive Proportion Test, §4.4.2): in a window of W = 512 samples (1,024 for a binary source), the test fails if a value appears at least C times, C being the smallest integer whose probability is $\leq 2^{-20}$.

Table 7. Health test cutoffs ($\alpha = 2^{-20}$).

Source and credited entropy H	RCT: C	APT: W	APT: C
jitterentropy, claim (1/3 bit)	61	512	449
Thermal noise of Lab 3 (4.0927 bits)	6	512	59
Ideal binary source (1 bit)	21	1,024	589

No sample enters the seed without passing these tests; at the first failure, the current seed is discarded.

7 Lessons from the labs

The NT-03 campaign applies the four tests to 22 datasets: healthy sources, design defects, environmental effects, failures and DRBG outputs. The "physical" sources are **simulated** from a model whose entropy is known, so that the measurement can be checked against the theory: 5.618 bits measured for 5.65 expected on thermal noise. Six lessons emerge.

1. **The tests do not see determinism.** An AES-256 stream with an all-zero key passes the IID tests with 7.87 bits per byte, and still gets 7.15 bits through the non-IID track, for a real entropy of zero. Rule: never assess a DRBG or hash output.
2. **IID does not mean high entropy.** A saturated converter passes the IID tests with only 1.92 bits of entropy out of 8.
3. **A single test is enough to reject the IID assumption.** A slowly read oscillator passes the chi-square and longest repeated substring tests, but a single permutation statistic out of 19 rejects it.
4. **The IID tests have a false rejection rate.** A truly IID source is rejected in 2 to 3 % of cases, because of the permutation threshold ($C \leq 5$ out of 10,000 for each of the 19 statistics); the standard then requires the non-IID track.
5. **Only the restart test sees some defects.** A source whose first samples are identical at every start-up looks normal in steady state; only test 3 rejects it.
6. **The conditioning margin is essential.** SHA-256 receiving exactly 256 bits of entropy outputs only 251.69 bits; with 320 input bits, the output has full entropy.

8 Conclusion

libsp80090b makes it possible to industrialise the qualification of entropy sources directly on their execution platform. It performs the four tests of SP 800-90B with the same results as the NIST tool, and provides the parameters of the hybrid generator: the number of seed samples N and the cutoffs of the RCT and APT health tests.

The measurement remains a check: the claimed entropy comes first from the model of the source, and a DRBG output fools every test. For an ESV validation submission, the NIST tool remains the reference.

9 References

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